THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Life Processes, Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** oxygen	Oxygen is released
2	**(b)** proteins	Pancreatic enzyme
3	**(c)** alveoli	Air sacs
4	**(a)** Correct	Low pressure
5	**(d)** artery	High pressure
6	**(a)** axon	Carries impulse away
7	**(c)** cerebrum	Fore-brain
8	**(b)** pancreas	Islets of Langerhans
9	**(c)** Incorrect — plants use hormones and cell changes	No nerves
10	**(b)** thyroxine	Thyroid

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Autotrophs make food from simple inorganic substances (e.g. green plants); heterotrophs depend on other organisms for food (e.g. animals).	2
12	Yeast cells; human muscle cells (during vigorous exercise)	1+1
13	Their pancreas does not secrete enough insulin, so blood sugar remains high.	2
14	Muscular tissue and glandular tissue	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) absorption of light by chlorophyll (ii) conversion of light energy to chemical energy and splitting of water (iii) reduction of CO_2 to carbohydrates	1+1+1
16	Bony skull (cranium) and a fluid-filled balloon (cerebrospinal fluid) act as shock absorbers; the brain is delicate and vital	2+1
17	Two destarched potted plants under bell jars, one with KOH (absorbs CO_2); after sunlight, iodine test: leaf with KOH does not turn blue-black	3
	OR: O_2: haemoglobin in RBCs carries it from lungs to tissues; CO_2: mainly dissolved in plasma, carried to lungs	1½+1½
18	Reflex: involuntary, quick, controlled by spinal cord; walking: voluntary, controlled by brain (cerebellum) and conscious thought	3
	OR: Auxin: cell elongation at shoot tip; gibberellin: stem growth; cytokinin: cell division, found in fruits and seeds	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) taking in and utilising food (ii) ingestion by pseudopodia → food vacuole → digestion → absorption → egestion (iii) Paramecium uses cilia to move food to a specific spot (cytostome)	1+3+1
	OR: (i) exhaled air tube (ii) exhaled air has more CO_2 (iii) CO_2 (iv) exhaled air: more CO_2, less O_2 (v) respiration	1+1+1+1+1
20	(i) dendrite, cell body, nucleus, axon, nerve ending (ii) electrical impulse travels dendrite → cell body → axon; chemicals at synapse start impulse in next neuron (iii) sensory: receptor to CNS; motor: CNS to effector	2+2+1
	OR: (i) reflex action (ii) spinal cord (iii) receptor → sensory neuron → relay neuron → motor neuron → effector muscle (iv) information also reaches the brain (v) sneezing/blinking	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Life Processes, Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** trachea	Prevent collapse
2	**(b)** Student A is correct	Villi
3	**(c)** night	Store CO_2 as acid
4	**(d)** urea	Removed by kidneys
5	**(a)** aorta	Largest artery
6	**(d)** hydrotropism	Water stimulus
7	**(a)** Student A is correct	Gaseous hormone
8	**(a)** pituitary	Growth hormone release
9	**(d)** growth hormone	Pituitary
10	**(a)** olfactory receptors	Nose

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water has much less dissolved oxygen than air, so aquatic animals breathe faster.	2
12	False — capillaries are one-cell thick to allow exchange of materials.	2
13	Detects stimuli and starts an impulse; gustatory receptors (tongue)	1+1
14	False — hunger is controlled by the hypothalamus; the cerebellum controls balance.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Removes nitrogenous wastes (urea), regulates water and salt balance, maintains blood pH	3
16	High blood sugar → pancreas secretes insulin → sugar falls → insulin release reduces. This prevents excess or deficiency.	3
17	Arteries: carry blood away from heart, thick elastic walls, no valves; veins: carry blood to heart, thin walls, have valves	3
	OR: Moistens food and contains salivary amylase that converts starch into sugar; secreted by salivary glands in the mouth	2+1
18	Impulse reaches the nerve ending, chemicals are released at the junction, muscle proteins change shape and arrangement causing contraction	3
	OR: Secreted in small amounts, carried by blood to target organs; growth hormone controls growth; insulin controls blood sugar	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) xylem vessels and tracheids; root cells take ions, water enters by osmosis; transpiration pull (ii) pressure created by water entering roots pushing water up (iii) at night when transpiration is low	3+1+1
	OR: (i) body → vena cava → RA → RV → pulmonary artery → lungs → pulmonary vein → LA → LV → aorta → body (ii) systole: contraction; diastole: relaxation	3+2
20	(i) cerebrum: thinking, senses, voluntary actions; cerebellum: balance and precision; medulla: involuntary; hypothalamus: hunger, temperature (ii) loss of balance and coordination	4+1
	OR: (i) chemicals coordinating plant growth: auxin, gibberellin, cytokinin, abscisic acid (ii) stem elongates towards light source	4+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Life Processes, Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** reflects green light	Absorbs red and blue
2	**(c)** Incorrect — they respire all the time	Day and night
3	**(a)** saprotrophic nutrition	Breakdown outside the body
4	**(c)** transpiration pull	Suction
5	**(a)** Both are correct	Normal 120/80
6	**(c)** heartbeat	Medulla/automatic
7	**(c)** Incorrect — endocrine glands secrete directly into blood	Ductless
8	**(b)** positive phototropism	Shoot
9	**(d)** central nervous system	CNS
10	**(a)** Student A is correct	Adrenaline raises it

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Creates an acidic medium for pepsin and kills microbes.	2
12	Blood would not clot; a wound would keep bleeding.	2
13	The shoot grows upward, away from gravity (negative geotropism).	2
14	It is a reflex action controlled by the spinal cord for protection.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water enters guard cells → they swell → pore opens for exchange; water lost → shrink → pore closes to reduce water loss	3
16	Reach only cells connected by nerves; after an impulse, the cell needs time to reset; cannot continuously generate impulses	3
17	Lack of O_2: pyruvate → lactic acid; accumulation of lactic acid causes cramps	2+1
	OR: Cellulose in grass takes longer to digest; cow (herbivore), tiger (carnivore)	2+1
18	Phototropism: response to light — shoots bend towards light; geotropism: response to gravity — roots grow downward	1½+1½
	OR: Cells change shape by changing water content; leaves fold and droop quickly without growth	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) diffusion of wastes through a semipermeable membrane (ii) nitrogenous wastes, excess salts (iii) both filter blood (iv) no reabsorption in artificial kidney (v) urea	1+1+1+1+1
	OR: (i) mouth: chewing, salivary amylase acts on starch; oesophagus: peristalsis; stomach: HCl, pepsin digests protein, mucus protects (ii) acid would damage the stomach lining causing ulcers	4+1
20	(i) goitre (ii) thyroid (iii) thyroxine (iv) regulates metabolism of carbohydrates, proteins and fats (v) iodine is needed to make thyroxine	1+1+1+1+1
	OR: (i) impulse vs hormone; fast vs slow; short vs long effect; nerves vs blood (ii) pancreas	4+1

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** blood capillaries	In Bowman's capsule
2	**(a)** lipase	Pancreatic/intestinal
3	**(c)** diffusion	Body surface
4	**(c)** right atrium	SA node
5	**(b)** guard cells	Turgor changes
6	**(d)** thyroid	Butterfly-shaped
7	**(d)** ovaries	Female hormone
8	**(c)** abscisic acid	ABA
9	**(d)** dendrite	Receives stimulus
10	**(a)** cerebellum	Precision of voluntary actions

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Glucose; amino acids; fatty acids + glycerol	1+1
12	They need high energy to maintain constant body temperature, so efficient oxygen supply is required.	2
13	Thyroxine; goitre	1+1
14	Organs (muscles/glands) that respond to impulses; e.g. arm muscles	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Helps absorption and upward movement of water and minerals; cools the plant; maintains turgidity	3
16	Testosterone in males and oestrogen in females cause secondary sexual characters, e.g. facial hair in boys, breast development in girls	1+2
17	(i) right atrium (ii) right ventricle (iii) left atrium	1+1+1
	OR: Large body; most cells are not in direct contact with environment; diffusion is too slow over large distances	3
18	Secretes growth hormone and others; controls the secretion of other endocrine glands	2+1
	OR: It releases releasing factors that stimulate the pituitary to secrete hormones (e.g. growth hormone)	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) synthesis of food from CO_2 and water using sunlight and chlorophyll; $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (ii) gas exchange, transpiration (iii) stored as starch	2+2+1
	OR: (i) inhalation: diaphragm flattens, ribs lift, chest cavity enlarges, air rushes in; exhalation: reverse (ii) provides time for O_2 absorption and CO_2 release	3+2
20	(i) pollen tube grows towards ovule (ii) tendrils coil around support (iii) tropic: directional growth; nastic: non-directional, not growth-dependent	2+2+1
	OR: (i) prepares the body for emergency (ii) adrenal glands (iii) heart rate increases, blood flows to muscles, breathing rate increases, digestion slows	1+1+3

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